A lower bound for the small-time cost of boundary control of the one-dimensional Schrödinger equation

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Abstract

We consider the one-dimensional Schrödinger equation on an interval of length L, driven by an L^2 Dirichlet control at one endpoint, with initial data measured in H^{-1}. We prove that its null-control norm cost satisfies $C_S(T,L) \geq c_L T^{7/2} \exp(L^2/(\pi T))$ for all sufficiently small T. Thus $\liminf_{T\downarrow 0} T \log C_S(T,L) \geq L^2/\pi$, excluding the proposed coefficient $L^2/4$. The proof uses one sine initial state: its exact control moments prescribe the zeros of an entire function, and Jensen's formula turns those zeros into the lower bound.

1 Introduction and main result

Boundary controllability asks how strongly a system must be driven at its boundary to reach a prescribed final state. For a Schrödinger equation on a segment, the small-time question concerns the growth of the required control norm as the available time tends to zero. It is closely related to the quantitative moment problem and entire-function methods developed for this setting; see Miller [1, Section 4].

Fix $L > 0$. We consider complex-valued transposition solutions of

$$\begin{cases} iy_t + y_{xx} = 0, & (t,x) \in (0,T) \times (0,L), \\ y(t,0) = h(t), \quad y(t,L) = 0, & 0 < t < T, \\ y(0,\cdot) = y_0 \in H^{-1}(0,L), \end{cases} \tag{1}$$

where $h \in L^2(0,T;\mathbb{C})$. We use H^{-1} as the continuous anti-dual of H_0^1, with a fixed standard Sobolev norm independent of T. The proof applies both to the derivative norm on H_0^1 and to its usual full Sobolev norm. Define the norm control cost by

$$C_S(T,L) = \sup_{\|y_0\|_{H^{-1}}=1} \inf\{\|h\|_{L^2(0,T)} : \ y(0) = y_0, \ y(T) = 0 \text{ in } (1)\}. \tag{2}$$

The infimum of an empty set and $\log(+\infty)$ are understood in the extended sense. In particular, C_S is a norm cost, rather than its square.

The lower coefficient $L^2/4$ appears in Lissy [2, Theorem 1.2, equation (8)]. For precisely the norm-cost setting (1)–(2), Lissy [3, Sections 3.1–3.2] conjectures

$$\lim_{T\downarrow 0} T \log C_S(T,L) = \frac{L^2}{4}. \tag{3}$$

The same work gives the upper bound $\limsup_{T\downarrow 0} T \log C_S(T,L) \leq 27^{1/4} L^2/4$ [3, Theorem 3.1]. Here we prove a strict improvement of the lower coefficient, which rules out (3).

Theorem 1. *For every fixed $L > 0$, there is a constant $c_L > 0$, independent of T, such that*

$$C_S(T, L) \geq c_L T^{7/2} \exp\left(\frac{L^2}{\pi T}\right), \qquad 0 < T \leq \frac{L^2}{2\pi}. \tag{4}$$

Consequently,

$$\liminf_{T \downarrow 0} T \log C_S(T, L) \geq \frac{L^2}{\pi} > \frac{L^2}{4}. \tag{5}$$

The constant c_L may depend on the fixed interval and Sobolev norm. The statement covers the full unit sphere in (2); a single unit sine state supplies its lower bound. It does not determine the sharp coefficient or establish that the limit exists.

The argument uses only necessary control moments. After shifting the first spectral frequency to zero, a centered transform of each control takes the value one at the origin and vanishes at all remaining quadratic spectral gaps. Its growth on a circle is bounded by the control norm and the time support. Jensen's formula compares these two facts. An exact factorial product then gives the exponential lower bound. Sections 2 and 3 derive the moments and the zero estimate; Section 4 chooses the radius and completes the proof.

2 Necessary moments for one unit initial state

We use the pairing linear in the state and conjugate-linear in the test function. For a transposition solution and a smooth homogeneous Dirichlet adjoint solution z, the endpoint identity is

$$\langle y(T), z(T) \rangle - \langle y_0, z(0) \rangle = i \int_0^T h(t) \overline{z_x(t, 0)}\, dt. \tag{6}$$

Here the adjoint solves $iz_t + z_{xx} = 0$ backwards from its terminal datum. For smooth states, differentiating $\int_0^L y\overline{z}\, dx$ gives $i[y_x\overline{z} - y\overline{z_x}]_0^L = ih\overline{z_x(t, 0)}$. Identity (6) is its transposed form; it requires no spatial trace of a rough state.

Let

$$\phi_n(x) = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}, \qquad \lambda_n = \frac{n^2\pi^2}{L^2}, \qquad b_n = \phi_n'(0) = \sqrt{\frac{2}{L}} \frac{n\pi}{L}. \tag{7}$$

These functions are orthonormal in $L^2(0, L)$ and satisfy $-\phi_n'' = \lambda_n \phi_n$ with zero endpoint values. Set

$$a_L = \|\phi_1\|_{H^{-1}}^{-1}, \qquad y_0 = a_L \phi_1, \qquad d_L = \frac{a_L}{b_1} > 0. \tag{8}$$

The norm in this definition is finite and nonzero: a smooth sine defines a continuous functional on H_0^1, and its pairing with itself is one. Thus y_0 has exactly unit H^{-1} norm. These quantities depend on L and the fixed Sobolev norm, but not on T.

For every control taking this state to zero, test (6) with $z(t, x) = e^{-i\lambda_n(t-T)}\phi_n(x)$. Since $z(0) = e^{i\lambda_n T}\phi_n$, we obtain

$$-a_L \delta_{n1} e^{-i\lambda_n T} = i b_n e^{-i\lambda_n T} \int_0^T h(t) e^{i\lambda_n t}\, dt.$$

Consequently every such control satisfies

$$\int_0^T h(t) e^{i\lambda_n t}\, dt = i d_L \delta_{n1}, \qquad n \geq 1. \tag{9}$$

Only smooth single-mode tests have been used. In particular, this necessary condition does not involve an infinite expansion of the controlled state.

3 The entire transform and its zeros

Normalize the nonzero moment and center the time interval by defining

$$v(t) = \frac{h(t)e^{i\lambda_1 t}}{id_L}, \qquad F(z) = \int_0^T v(t)e^{iz(t-T/2)}\,dt, \qquad z \in \mathbb{C}. \tag{10}$$

Cauchy–Schwarz gives $\|v\|_1 \le \sqrt{T}\,\|v\|_2 = \sqrt{T}\,\|h\|_2/d_L$. The transform is entire. Indeed, on $|z| \le A$, its exponential series can be integrated term by term, with remainder at most

$$\|v\|_1 \sum_{k>K} \frac{(AT/2)^k}{k!} \longrightarrow 0.$$

The resulting power series has infinite radius of convergence. Also,

$$|F(z)| \le K_T e^{(T/2)|\operatorname{Im} z|}, \qquad K_T = \frac{\sqrt{T}\,\|h\|_2}{d_L}. \tag{11}$$

The factor $T/2$ is the reason for centering the interval.

The moments (9) imply

$$F(0) = 1, \qquad F(\mu_n) = 0 \quad (n \ge 2), \qquad \mu_n = \lambda_n - \lambda_1 = \frac{\pi^2}{L^2}(n^2-1). \tag{12}$$

Thus F is nonzero and has distinct positive prescribed zeros. It may have other zeros, which only strengthen the estimate below.

Let $r > 0$ be a radius with no zero on its circle. Jensen's formula at zero, with zeros z_j counted according to multiplicity, reads

$$\sum_{|z_j|<r} \log\frac{r}{|z_j|} = \frac{1}{2\pi}\int_0^{2\pi} \log|F(re^{i\theta})|\,d\theta,$$

because $F(0) = 1$. The sum is finite, since zeros of a nonzero entire function are isolated. Each summand is nonnegative. Taking only the prescribed zeros and applying (11), we find

$$\sum_{\mu_n<r} \log\frac{r}{\mu_n} \le \log K_T + \frac{Tr}{\pi},$$

where $\int_0^{2\pi} |\sin\theta|\,d\theta = 4$ fixes the constant. For any $R > \mu_N$, with an integer $N \ge 2$, it follows that

$$\log K_T \ge \sum_{n=2}^{N} \log\frac{R}{\mu_n} - \frac{TR}{\pi}. \tag{13}$$

To include radii whose circles contain additional zeros, choose zero-free radii increasing to R, all larger than μ_N. Such radii exist because the zero set is countable. Pass to the limit only in the selected finite sum. This proves (13) at every required R.

4 The lower bound

Choose $R = \pi^2 N^2/L^2$, so $\mu_N < R$. The prescribed-zero contribution is

$$S_N := \sum_{n=2}^{N} \log\frac{R}{\mu_n} = 2(N-1)\log N - \log\frac{(N-1)!(N+1)!}{2}. \tag{14}$$

Here we used $\prod_{n=2}^{N}(n^2-1) = (N-1)!(N+1)!/2$. Since log is increasing,

$$\log N! \le \int_1^N \log x\, dx + \log N = N\log N - N + 1 + \log N.$$

Together with $(N-1)!(N+1)! = (N!)^2(1+1/N)$, this gives, for every $N \ge 2$,

$$S_N \ge 2N - 4\log N - 2 + \log 2 - \log(1+1/N) \ge 2N - 4\log N - 2. \tag{15}$$

Substituting into (13) and recalling (11), we obtain

$$\|h\|_2 \ge \frac{d_L}{\sqrt{T}}\, N^{-4} e^{-2} \exp\!\left(2N - \frac{\pi T}{L^2} N^2\right), \qquad N \ge 2. \tag{16}$$

This holds for every admissible control of the fixed unit state and every integer $N \ge 2$.

Proof of Theorem 1. For $0 < T \le L^2/(2\pi)$, set $X = L^2/(\pi T) \ge 2$, $N = \lfloor X \rfloor$ and $a = 1/X \le 1/2$. Completing the square gives

$$2N - aN^2 = X - a(N-X)^2 \ge X - \tfrac{1}{2}, \qquad N^{-4} \ge X^{-4} = \left(\frac{\pi T}{L^2}\right)^4.$$

Therefore (16) implies

$$\|h\|_2 \ge c_L T^{7/2} e^{L^2/(\pi T)}, \qquad c_L = d_L \pi^4 L^{-8} e^{-5/2} > 0. \tag{17}$$

The constant is independent of T and of the control. Taking the infimum over all controls for the unit state (8), and then the supremum over the whole unit sphere, proves (4). Taking the logarithm, multiplying by T, and using $T\log T \to 0$ proves (5). The strict inequality follows from $\pi < 4$. □

AI-use statement. OpenAI Codex was used for literature searches, proof development, calculation checks, and drafting this manuscript. The author is responsible for its mathematical claims.

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